

Metapopulation Modelling of Epidemic Peak Timing in Major Russian Cities with Transport Flows

Iskandar Akhmetov

Abstract

This study investigates the influence of inter-city transport flows on the timing of epidemic peaks of acute respiratory viral infections (ARVI) in 12 major Russian cities. Analysis of weekly incidence data from 1985 to 2016 reveals statistically significant correlations in disease dynamics between cities. A metapopulation SIR model incorporating an expert-based transport flow matrix is constructed. The model is calibrated on 2009 data and validated on independent years (2007, 2011, 2013, 2015). Comparison with a model without transport connections demonstrates a consistent reduction in peak timing prediction error: the median absolute error decreases from 4.79 to 1.14 weeks. The results confirm that accounting for transport flows substantially improves the accuracy of epidemic peak timing forecasts.

Keywords: metapopulation model, SIR, transport flows, epidemic forecasting, ARVI, Russia

1. Introduction

Acute respiratory viral infections and influenza remain among the most widespread infectious diseases, causing millions of severe cases and hundreds of thousands of deaths annually [1]. For planning anti-epidemic measures, accurate estimates of not only the height of epidemic waves but also the timing of incidence peaks are essential.

Classical approaches to modelling influenza spread between cities date back to the work of Baroyan and Rvachev [2], in which local SEIR models were coupled through linear migration flows. Under modern conditions, the

*Corresponding author

accuracy of such models declines due to changes in herd immunity, contact structure, and the quality of population movement data [3].

In the study by Leonenko and Novoselova [3], based on Russian data from 2000–2015, it was shown that standard coupled models can satisfactorily estimate peak height; however, reliable prediction of the *day (week) of the peak* remains challenging. The authors noted that improving accuracy requires higher-quality data on inter-city migration flows.

The aim of the present work is to test the hypothesis that incorporating a realistic structure of transport flows between major Russian cities can substantially improve the prediction of ARVI epidemic peak timing. To this end:

1. Correlation analysis of incidence between 12 cities was performed.
2. A metapopulation SIR model with an expert-based transport flow matrix was constructed.
3. The model was calibrated and subjected to multi-year validation.
4. A direct comparison of prediction accuracy with a model without transport connections was conducted.

2. Data and Preliminary Analysis

Weekly ARVI incidence data for 12 Russian cities covering the period 1985–2016 were used: Moscow, Saint Petersburg, Novosibirsk, Yekaterinburg, Nizhny Novgorod, Samara, Omsk, Kazan, Chelyabinsk, Ufa, Perm, and Rostov-on-Don.

After data cleaning and construction of a panel structure, pairwise Pearson correlation coefficients were calculated. Sixty-six statistically significant pairs ($p < 0.01$) were identified, with correlation coefficients predominantly in the range 0.65–0.88. The strongest synchronisation is observed between cities connected by intensive transport corridors (Moscow–Saint Petersburg, Ural cities, Novosibirsk–Omsk).

These findings confirm that epidemic waves in different cities are not independent and that transport flows may serve as one of the mechanisms of their synchronisation.

3. Model

3.1. Metapopulation SIR Model

A system of 12 coupled populations is considered. For each city i , the dynamics are described by the equations:

$$\frac{dS_i}{dt} = -\beta \frac{I_i}{N_i} S_i + \sum_{j \neq i} (M_{ji} S_j - M_{ij} S_i), \quad (1)$$

$$\frac{dI_i}{dt} = \beta \frac{I_i}{N_i} S_i - \gamma I_i + \sum_{j \neq i} (M_{ji} I_j - M_{ij} I_i), \quad (2)$$

$$\frac{dR_i}{dt} = \gamma I_i + \sum_{j \neq i} (M_{ji} R_j - M_{ij} R_i). \quad (3)$$

where M_{ij} is the fraction of the population of city i moving to city j per unit time.

3.2. Transport Flow Matrix

Instead of a pure gravity model, an expert-based matrix of relative flows was used, reflecting the real structure of passenger traffic in Russia:

- Moscow as the main transport hub;
- intensive Moscow–Saint Petersburg corridor;
- Trans-Siberian route;
- Ural cluster (Yekaterinburg–Chelyabinsk–Perm–Ufa).

The matrix is scaled by a single calibratable parameter `mobility_scale`.

3.3. Calibration and Validation

Parameters β , γ , `mobility_scale`, and the size of the initial infection seed were calibrated on 2009 data using the Nelder–Mead method by minimising the mean absolute error (MAE) of peak timing.

Validation was performed on independent years: 2007, 2011, 2013, and 2015. For each year, two model versions were compared — with transport flows and without them (near-zero `mobility_scale`).

4. Results

4.1. Prediction Quality

Summary results for absolute errors in peak timing prediction (all cities and years):

Table 1: Absolute peak timing errors across all years and cities

Model	Median error	Mean	Bootstrap 95% CI of median
With transport	1.14	1.32	[0.79; 1.43]
Without transport	4.79	4.35	[3.29; 5.43]

The median error of the model with transport is approximately 4.2 times lower. The confidence intervals of the medians do not overlap, indicating a statistically significant difference.

4.2. Robustness Across Years

The advantage of the transport-aware model is observed in all considered years. The most pronounced effect is seen in 2009, 2011, and 2015. In 2015, the median error of the transport-aware model was approximately 0.6 weeks.

4.3. Quality Metrics

Additionally, RMSE, Hit Rate (± 1 and ± 2 weeks), and Pearson and Spearman correlation coefficients between observed and modelled peak times were calculated. On all metrics, the transport-aware model outperforms the model without transport connections.

5. Discussion

The obtained results differ substantially from the conclusions of [3], in which coupled models failed to reliably improve peak timing forecasts. Possible reasons for the success of the present work include:

- use of an expert matrix that better reflects the real flow structure than a simple gravity model;
- direct calibration on peak timing (rather than on the shape of the entire epidemic curve);

- a more homogeneous set of 12 large cities.

The main limitations of the study are:

- the flow matrix is expert-based rather than constructed from real railway and air traffic data;
- transmission parameters are assumed identical for all cities;
- seasonal forcing $\beta(t)$ is not explicitly included.

Despite these limitations, the model demonstrates a robust and quantitatively significant advantage when transport connections are taken into account.

6. Conclusion

The study shows that:

1. ARVI incidence dynamics in major Russian cities are strongly correlated.
2. A metapopulation SIR model with a realistic transport flow matrix substantially improves the prediction of epidemic peak timing.
3. Accounting for transport reduces the median absolute error from 4.79 to 1.14 weeks.

These results confirm the practical value of incorporating transport flows into epidemic forecasting models for the territory of Russia.

References

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